Suchitra Anand Gunisetty

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Professional Summary

6 years of diversified experience in Software Design Development. Experience as Big Data Engineer solving business use cases for several clients. Experience in the field of software with expertise in backend applications.

Relevant Coursework: Design and Analysis of Algorithm, Advanced Software Engineering, Web Stack, Mobile Computing, Data Science, Database Systems, Foundations in Artificial Intelligence.

Proficient in AWS services (S3, Glue, Lambda, RDS, Redshift) with expertise in building and managing cloud-based data solutions, leveraging serverless architecture for data pipelines, Hands-on experience in PyCharm IDE for writing and debugging Python and PySpark code.

Strong knowledge in CI/CD pipelines using AWS CodeBuild, CodeDeploy, and other DevOps tools to streamline deployment processes and improve development efficiency.

Experienced in using PySpark for distributed data processing, optimizing performance with partitioning, caching, and custom UDFs, enhancing the speed and reliability of data workflows.

Expertise in Snowflake data warehousing, along with AWS Glue and Redshift, to architect and optimize large-scale data lakes and lakeshores for structured and semi-structured data.

Technical Skills

Languages:	Python (Tensorflow, OpenCV, Numpy, Pandas, Scipy, Matplotlib), PySpark, Scala, C++, Shell Scripting (Bash)
Web Technologies:	RESTful APIs, HTML, CSS, Flask, Spring Framework, Hibernate
Databases:	SQL, MySQL, Amazon RDS, Amazon DynamoDB, MongoDB, PostgreSQL
Developer Tools:	Linux, Git, VS Code, Eclipse, Pycharm, Android Studio, High-Performance Computing Clusters
Cloud Technologies:	AWS (VPC, CloudFormation, Code Deploy, Lambda, S3, Glue, API Gateway, CloudWatch, IAM), On-Premises
Visualization Tools:	Power BI, Tableau, Excel

Education

Master of Science, Computer Science Jan. 2022 – Dec. 2023
Jessup University *California, USA*
Relevant Coursework: Design and Analysis of Algorithm, Advanced Software Engineering, Web Stack, Mobile Computing, Data Science, Database Systems, Foundations in Artificial Intelligence.

Bachelor of Technology, Computer Science Aug. 2016 – May. 2020
Jawaharlal Nehru Technological University *Hyderabad, India*
Relevant Coursework: Data Structures, Object-Oriented Programming with CPP and JAVA, Programming Languages, Computer Networking, Operating Systems, Database Management

Relevant Experience

Sr. Data Engineer April 2024 - Present
Navient *Fishers, IN*

- Involved in converting Hive/SQL queries into Spark transformations using Spark RDDs, Python and Scala.
- Developed Spark transformations using PySpark to handle large datasets.
- Implemented Snowflake as the central data warehouse to store, process, and analyze large datasets from various sources.
- Optimized data transformations using RDDs and DataFrames for faster processing.
- Leveraged AWS Glue for building ETL pipelines, integrated with S3 and Lambda for batch and real-time data processing.
- Used PyCharm for building and debugging PySpark code, ensuring optimized code delivery.

Hadoop Developer/ Admin November 2023 - April 2024
Navient *Fishers, IN*

- Designed, coded, and implemented efficient Hadoop ETL solutions for the Enterprise Data Warehouse, leveraging Python to modernize the existing ETL framework.
- Provided 24/7 on-call production support for the Cloudera CDP environment, resolving job failures and communicating effectively with users about EDW delays and changes.
- Implemented data pipelines using HDFS for storage, Hive for data warehousing, and Impala for high-performance querying.
- Worked in sprints, utilizing Talend Big Data components such as Hadoop, S3 Buckets, and AWS services for Redshift.
- Developed job workflow in Oozie to automate the tasks of loading the data into HDFS and a few other Hive jobs.
- Involved in migrating an on-prem Hadoop system to Google Cloud Platform (GCP).

Data Engineer January 2023 - August 2023
Global Logic/ Capital Group *San Antonio, TX*

- Designed data pipelines leveraging Snowflake for cloud-based data processing and storage, integrated with AWS Glue and S3.
- Engineered multiple Glue jobs to consume data from the S3 standard layer, loading it into the Postgres master table.
- Developed and deployed Lambda functions for a serverless data pipeline using SQS services, orchestrated on Airflow.
- Developed scalable data pipelines and ETL processes using Scala, Python, and Spark to analyze large datasets, optimizing performance through efficient data handling.
- Integrated CI/CD pipelines with GCP services such as Cloud Dataflow, BigQuery, and Google Kubernetes Engine (GKE) to ensure smooth and automated releases of data processing applications.
- Automated the provisioning of infrastructure using Infrastructure-as-Code (IaC) tools like Terraform, integrated into the CI/CD pipelines to manage GCP resources.
- Regularly reviewed and updated GCP service configurations (IAM, KMS, DLP) to ensure alignment with changing security requirements and evolving industry standards.
- Applied PyCharm for development and debugging of PySpark jobs, ensuring seamless execution and debugging.
- Configured resource-level IAM policies for GCP services like BigQuery, Cloud Storage, and Pub/Sub to maintain data security and access control.
- Built scalable data pipelines using PySpark and Spark, optimizing for large data processing.

Graduate Assistant May 2022 - Dec 2022
Department of Computer Science *Jessup University-San Jose, CA*

- Built and validated a machine learning model to study human behavior and mentality towards Covid-19. Performed data science tasks including data collection, cleaning, imputation, visualization, statistical testing, and modeling using scikit-learn, pandas, scipy, and matplotlib. Achieved an accuracy of over 78% through the use of an ensemble model.
- Enhanced existing research by applying a neural network model to solve partial derivative equations (PDEs) known as Minimal Surface using tensorflow, numpy, and Linux-based high-performance computing (HPC) clusters.

Associate Data Engineer July 2019 - November 2021
TATA AIG *Hyderabad, India*

- Built data engineering pipelines using Scala in AWS for financial data analysis.
- Refactored Python codes to Pyspark to utilize Spark parallelization, emphasizing strong Scala programming skills.
- Designed data pipelines using AWS Glue, Redshift, and S3 for large-scale data integration. Implemented AWS RDS and Lambda for serverless data operations.
- Optimized Spark job performance using advanced techniques like broadcast variables, dynamic allocation, and partitioning.
- Implemented batch and streaming ingestion of data, optimizing Spark performance using broadcast variables, dynamic allocation, partitioning, and building custom Spark UDFs.
- Integrated Glue with other AWS services like S3, Redshift and RDS to facilitate data movement transformation

Selected Projects | More Projects on Github

Motel Check-in Bot | Python, Selenium, AWS Lambda

- Automated the check-in process at a local Motel using Python and AWS Lambda.
- Deployed a Linux virtual machine on AWS to ensure the program could run efficiently and automatically check in at a given time
- Used Cron Job to schedule the program to execute automatically at a predetermined time to secure a reservation.

Data Mining Approach to detect Website Phishing

- Devised and implemented a Data Mining Methodology targeting website phishing, employing features like web-page ranking, validity status, URL characteristics, and presence of special characters.
- Employed classification algorithms such as KNN, Logistic Regression, and SVM to construct a predictive model, enabling proactive detection of phishing URLs and mitigating potential cyber-attacks.